knew how to build the cluster test stand as much as this one. The difference in cost was estimated to be around Rs 1 crore; not too big an amount. I told this to Dhawan, who thought it was unnecessary at this stage since we were still experimenting with our first engine, and a cluster engine configuration for our rockets was an idea light years away.

'Let's first learn to walk,' Dhawan said. 'After that we will run.'

Nobody argues with Dhawan. I kept silent, but the thought of upgrading the principal test stand was tickling me. I have worked around bureaucratic barricades and have bent a few rules when I was convinced they were standing in the way of better results. This I did, also, when I was convinced that my visionary bosses like Sarabhai and Dhawan, who had no axe to grind, would soon see reason in the exceptions to the rule and approve of them.

Now I was planning to get the foundation laid for an unapproved cluster-engine test stand, with the confidence that I can convince the chairman sooner than later. After all, once the foundation is laid at 4m x 4m, the approval for a bigger test stand would be of no use.

I was just in time when R D John, the chief engineer of construction, was doing the foundation layout on the ground. I casually told John to make it 5m x 5m.

'But the approved plan is 4m x 4m,' said John.

'Yes, but don't worry, I will get approval for 5m x 5m.'

John wasn't convinced. He said the change would entail an additional expense of Rs 1 crore, and that I have to get a note approving the change.

Getting Rs 1 crore, I calculated, was not a big deal. I could put it under 'unforeseen expenses' or do some re-appropriation of fund since I was convinced that a four-engine test pad would be an asset for future. And I was sure that I could explain things to Dhawan later. I convinced John to do the ground markings as I wanted, assuring him that I would get the note soon.

Incidentally, on the day the marking pegs were driven into ground at the test site, Dhawan was in Trivandrum, and his flight back to Bangalore was cancelled. Not one to relax, the ISRO chairman said he would use the time